



BLITTA – TOGO SOLAR POWER PLANT

Document Prepared by ELEMENT 6 - FZCO

ON BEHALF OF CLIMATEPARTNER GmbH



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1 PROJECT DETAILS

1.1 Summary Description of the Project

Togo, which imports more than half of its energy from Nigeria and Ghana, is banking on solar power to develop access to electricity for its eight million residents. The Togolese Government understands that the key to economic development and growth depends on access to electricity but face the challenge of lack of financial resources. The electrification strategy of Togo has set ambitious targets to achieve universal access by 2030.

The Blitta - Togo solar PV plant is one of the largest in West Africa and the first renewable energy facility in the country^{1,2}. This plant, which is located in Blitta, will supply reliable, clean electricity to the national grid which then supplies to hundreds of thousands of homes and businesses in the country and save more than one million tonnes of CO₂ emissions through its lifetime.

The current capacity of the plant is 50 MW. An additional 20 MW expansion is expected to be operational by December 2023 bringing the total to 70 MW. The village of Blitta-Losso is part of the canton of Blitta village and is surrounded from north to south and from east to west by the villages of Blitta kotokoli, Boufouli Boko Losso, Doufouli and Oranyi. The Blitta area has an average monthly direct normal radiation of 5.14 kWh/m²/day and an average annual solar irradiation of 1,876 kWh/m². The coordinates of the project location are 8° 21'52.33"N, 1° 1'4.42"E. The project will advance the Togolese national clean energy strategy to increase the share of renewables in electrification to 50 per cent by 2025 and 100 per cent by 2030.

The project activity, which has a lifetime of 25 years, is expected to produce an average of 107,027 MWh of electricity annually. The total estimated emission reductions of the project are 490,607 tCO₂e during the first crediting period. The ex-ante emission reduction calculation shows that the project will avoid an average of 61,326 tCO₂e annually. The baseline scenario is the electricity generated and delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources. The project will not only reduce GHG emissions but is expected to achieve many social benefits such as improving social services to targeted social groups and generating job opportunities. Many job opportunities have already been created for the local people that are intended to promote sustained, inclusive, and sustainable economic growth. The project is also helped ensure that the local stakeholders acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, and gender equality.

¹ <https://www.irena.org/newsroom/pressreleases/2021/Jun/Togo-Inaugurates-50MW-Solar-Plant-Financed-Under-IRENA-ADFD-Facility>

² <https://www.mofaic.gov.ae/en/mediahub/news/2021/6/24/24-06-2021-uae-togo>

1.2 Sectoral Scope and Project Type

Sectoral Scope: 1, Energy industries (renewable -/non-renewable sources)

Type: Renewable energy projects (Large Scale)

1.3 Project Eligibility

According to the latest VCS Standard, version 4.3, the project is eligible under the scope of the VCS Program because:

- It results in CO₂ emission reductions, one of the seven greenhouse gases under the Kyoto Protocol
- It consists of grid-connected electricity generation using (large scale) solar power plant built in a Least Developed Country.

1.4 Project Design

The project has been designed to include a single installation of an activity.

- ☒ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☐ The project is a grouped project

Eligibility Criteria

The proposed project is not a group project activity. Thus, this section is not applicable.

1.5 Project Proponent

Organization name	AMEA SOLAR TOGO SAU
Contact person	Basma Bentaher
Title	Senior Director – Project Development
Address	Marina Plaza Offices 3301/3302/3303 Level 33 Dubai Marina PO Box 37669 Dubai United Arab Emirates

Telephone	+971 4 310 7014
Email	Basma.Bentaher@ameapower.com

Organization name	ClimatePartner GmbH
Contact person	Dr. Sascha Lafeld
Title	Head of Carbon Offsets and Green Energy Services
Address	St.-Martin-Str. 59, D-81669 Munich, Germany
Telephone	+49 89 1222 8750
Email	sascha.lafeld@climatepartner.com

1.6 Other Entities Involved in the Project

Organization name	Element 6 – FZCO
Role in the project	Consultant
Contact person	Amer Arafat
Title	Partner
Address	Technohub 1, DTEC, Dubai Silicon Oasis, Dubai, UAE
Telephone	+971 52 938 9590
Email	amer@element6.cc

1.7 Ownership

The project activity is fully owned by the project proponent AMEA SOLAR TOGO SAU. The ownership of the project can be proven through:

- Certificate of Completion signed by the Togolese Republic
- Power Purchase Agreement (PPA) signed by AMEA SOLAR TOGO SAU and the Togolese Republic
- Declaration of Conformity signed by the Togolese Republic

- Entry into Force of the Concession Agreement signed by AMEA SOLAR TOGO SAU and the Togolese Republic
- Back Energization of the PV Plant Document signed by the Togolese Republic, the Electricity Off-taker and AMEA SOLAR TOGO SAU.

1.8 Project Start Date

The project's COD (Commercial Operation Date) was 12-June-2021, which is the date on which the project started generating emission reductions. According to the Declaration of Conformity signed by The Minister Delegate to the President of the Republic in charge of Energy and Mines, and the Minister of Economy and Finance, the technical tests for the commissioning of the power plant took place over the period from March 15 to June 05, 2021, including performance tests. The commissioning date of Phase 1 was on 12-June-2021 while the commissioning date of Phase 2 is planned for 01-December-2023.

1.9 Project Crediting Period

The project's crediting period is 7 years renewable.

The first crediting period starts from 12-June-2021 and lasts till 11-June-2028.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☒ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☐ >1,000,000 tCO₂e/year

Project Scale	
Project	√
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
------	--

2021	39,985
2022	70,822
2023	70,468
2024	70,308
2025	69,765
2026	69,416
2027	69,069
2028	30,774
Total estimated ERs	490,607
Total number of crediting years	7
Average annual ERs	61,326

1.11 Description of the Project Activity

The project activity consists of setting up a 70 MW solar PV plant in Blitta-Lasso in Togo, with 50 MW power already installed and a planned expansion of 20 MW. The land is 117 ha large and is made available by the Togolese Republic. The Blitta area has an average monthly direct normal radiation of 5.14 kWh/m²/day and an average annual solar irradiation of 1,876 kWh/m².

Togo recognizes that access to electricity is an essential element for its economic development and inclusive growth policy and is one of the major pillars that will enable the implementation of its new National Development Plan. Their goal is to ensure universal access for all Togolese by 2030, with a 100% access rate over the next ten years that will be achieved through an intelligent combination of network expansion and off-grid technologies (mini-grid solar kits).

Table 1 - Specifications of the PV Panels

Manufacturer: Jinko Solar

Power installed on the ground: 50 MWp with 20 MWp planned expansion

Operating Temperature (°C): -40°C~+85°C

Power Tolerance: 0~+3%

The technical specifications of each type of the modules are shown in the tables below:

Module Type	JKM390M-72-V
Maximum Power (P _{max})	390Wp
Open-circuit Voltage (V _{oc})	49.3V
Maximum Power Current (I _{mp})	9.49A

Module Efficiency STC (%)	19.67%
No of modules	55,242

Module Type	JKM395M-72-V
Maximum Power (P_{max})	395Wp
Open-circuit Voltage (V_{oc})	49.5V
Maximum Power Current (I_{mp})	9.55A
Module Efficiency STC (%)	19.92%
No of modules	66,300

Module Type	JKM400M-72-V
Maximum Power (P_{max})	400Wp
Open-circuit Voltage (V_{oc})	49.8V
Maximum Power Current (I_{mp})	9.60A
Module Efficiency STC (%)	20.17%
No of modules	5,940

Module Type	LR5-72HBD-545M
Maximum Power (P_{max})	545Wp, Bifacial
Open-circuit Voltage (V_{oc})	49.8V
Maximum Power Current (I_{mp})	13.12A
Module Efficiency STC (%)	21.3%
No of modules	36,708

Table 2 - Specifications of the Inverters

Make and model	Sungrow SG250HX- 250KTL – H1 String or equivalent
Nominal apparent power	250 kVA @30°C/ 225 kVA @40°C/200 kVA @50°C
Rated current (I_n)	126.3A @40°C, 121.3A @45°C, 108.3A @50°C
Maximum apparent power	200 × 250,000 VA
Rated output voltage	800 V
Type of connection	Triphase
No. of inverters	200

Table 3 -Specifications of the Inverter Flow Transformers

Model	External mineral oil (STS-6000K) or equivalent
Rated power	6 × 6000 kVA @40°C/5400 kVA @50°C and 2 x 8000 kVA @40°C
Primary voltage	800 V
Secondary voltage	33 kV
No-load losses	6.25% (±10%)
Losses at rated current	<4.8 kW (IEC 60076-1) (IEC 62271-202, EN 50588-1, IEC 61429-1)

1.12 Project Location

The Government of Togo (Ministry of Mining and Energy) signed a Power Purchase Agreement with the solar developer, AMEA SOLAR TOGO SAU, aiming to develop a 70 MW solar power plant in Togo. The coordinates of the project location are 8°21'52.33"N, 1°1'4.42"E and covers a land area of 117ha. Additionally, the PV plant will be connected to the national grids 161 kV transmission line located close to the site³. The Blitta area has an average monthly direct normal radiation of 5.14 kWh/m²/day and an average annual solar irradiation of 1,876 kWh/m².

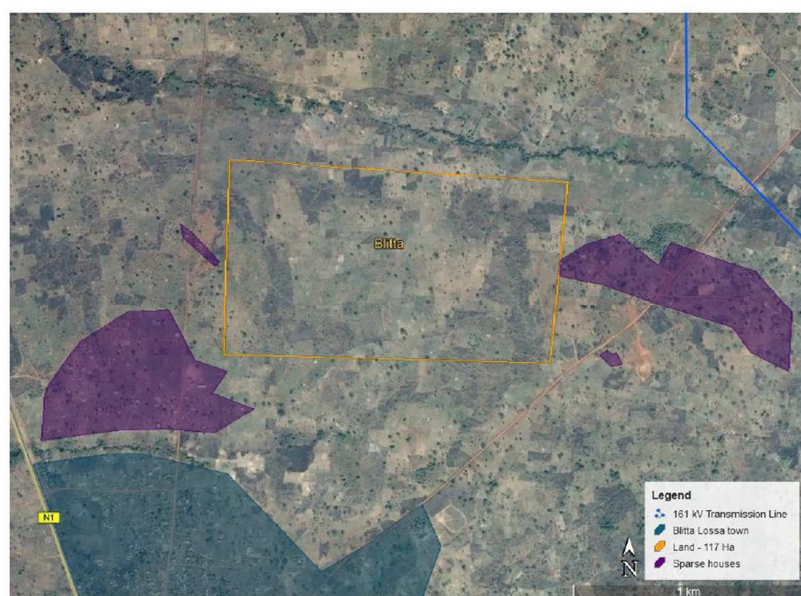


Figure 1 - Site location for the 70MW PV Power Plant

³ <https://energie.gouv.tg/projet/construction-dune-centrale-solaire-de-50-mwc-a-blitta/>

1.13 Conditions Prior to Project Initiation

In accordance with methodology ACM0002 Version 20 the project activity is the installation of a new grid-connected renewable power plant (Greenfield).

The baseline scenario is defined as electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system.”

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Togolese government created the Togolese Rural Electrification and Renewable Energies Agency (AT2ER) (Presidential decree N ° 2016 - 064 / PR of May 11th, 2016), whose mission is to promote renewable energy in Togo. The creation of the AT2ER was accompanied in July 2018 by the announcement of the Law 2018-010 08/08/2018 on the promotion of electricity production based on renewable energy sources in Togo. This resulted in creation of a specific legal framework to respond adequately to the challenges related to the production, marketing and use of electricity generated from renewable energy in Togo.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project is neither registered nor seeking registration under any other GHG programs.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes ☒ No.

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

1.17 Sustainable Development Contributions

According to World Bank data from 2020, around 54% of the Togolese population had access to electricity⁴. In order to meet the needs of the Togolese people, the Government of Togo has committed to an ambitious policy on energy access that is to achieve a 100% electricity access rate in the country by 2030⁵. In this context, the country has subscribed at regional and international level to many commitments in this sector and which require that the share of renewable energy to be increased to 50% in the national energy mix by 2025⁶. This is in accordance with Goal 7 of the United Nations Sustainable Development Goals, which aims to ensure access for all to reliable, sustainable, and modern energy services and at affordable costs.

In addition, AMEA SOLAR TOGO SAU has put in place several CSR activities across essential sectors. The activities are developed in consultation with the community near the Britta PV plant and are in accordance with the following Sustainable Development Goals (SDGs):



⁴ <https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS?locations=TG>

^{5,6} https://unfccc.int/sites/default/files/NDC/2022-06/CDN%20Revis%C3%A9es_Togo_Document%20int%C3%A9rimaire_rv_11%2010%2021.pdf



SDG	SDG Overview	Action Taken	Monitoring Parameters
3 Good Health and Wellbeing	Ensure healthy lives and promote well-being for all at all ages	- Construction of a clinic in the village of Blitta for curative and preventive service to the members of the local community. - Distributed essential food and hygiene products to 2,200 families as a response to the lockdown in Togo.	3.1. Number of patients visiting the newly constructed clinic (700 patients per year)
4 Quality Education	Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all	- Construction of three primary schools (Blitta, Tomegbe and Kokonade) - Renovation of four primary schools (Blitta Centre, Carrefour, Finade and Tchangaidé). - Provision of necessary furniture and educational equipment - Internship program for 36 students from Lomé Technical University	4.1. Number of students going to primary schools (750 students per year)
7 Affordable and Clean Energy	Ensure access to affordable, reliable, sustainable, and modern energy for all	Construction and commissioning of the Blitta Solar	7.1. Total clean energy generated from the Blitta Solar Plant. Measured in MWh/year. (Around 88,000 MWh for 2022, increasing to around 99,000 MWh in 2023 and

			around 120,000 MWh in 2024 onwards)	
8	Decent Work and Economic Growth	Promote sustained, inclusive and sustainable economic growth, full and productive employment, and decent work for all	Temporary jobs created during the construction of the plant and permanent jobs created for the operation and maintenance of the plant	8.1. Number of employees working in the operation and maintenance activities of the plant (10 employees)
13	Climate Action	Take urgent action to combat climate change and its impacts	Reduction of total carbon dioxide emissions generated.	13.1. Amount of annual emission reductions achieved from the project activity. Measured in (tCO ₂ e) ((Around 50,400 tCO ₂ e for 2022, increasing to around 56,700 tCO ₂ e in 2023 and around 68,700 tCO ₂ e MWh in 2024 onwards)

1.18 Additional Information Relevant to the Project

Leakage Management

As per approved methodology ACM0002 (Version 20.0), no leakage is considered for the project activity.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

No further information to be provided.

2 SAFEGUARDS

2.1 No Net Harm

The activities during construction, operation, and decommissioning of the project have been analysed to identify corresponding environmental and social impacts and assess their significance. Mitigation measures have been identified in the Environmental and Social Management Plan (ESMP) for negative impacts where institutional responsibilities for implementation and follow ups have been presented. The socioeconomic benefits of the project therefore outweigh the potential negative environmental and social impacts.

2.2 Local Stakeholder Consultation

A Local Stakeholder Consultation process has been conducted in accordance with the VERRA-Verified Carbon Standard (VCS) requirements, towards the successful registration of emission reduction projects/programs aiming to generate carbon credits for the project activity. The details of the LSC are as follows:

Venue: CICB (Centre International de Conférence de Blitta)

Date: Tuesday, 5-July-2022

Time: 08:00 AM GMT

The objective of the Stakeholders' Consultation event was to take due diligence of all comments, questions, and opinions in an open and transparent manner towards the projects' economic, social, and environmental impacts, and sustainability. It was also to raise the stakeholders' awareness level about the procedure to register such projects under the VCS.

The Local Stakeholder Consultation report (See Appendix A) is devised to provide the outcomes of the stakeholder consultation, and to ensure the transparency of the process through documentation.

The invitees were sent the invitation below to attend the stakeholders' consultation event. They were also requested to submit their written comments, should they have any, via email to Amer Arafat at info@element6.cc.



Lomé, le 20 juin 2022

Monsieur M. BOUKARI BATOSSA
Préfet de la préfecture de Bliha,
BP: 01 Bliha, Gare
République du Togo

Référence : BLT_009_06_22/PREFET/AMEA

Concernant : Projet AMEA - Centrale Solaire de 70 MWc à Bliha.

Objet : Invitation des parties prenantes locales à la consultation publique

Monsieur le **Préfet**,

Dans le cadre de la valorisation du projet de centrale solaire de Bliha pour l'obtention des crédits carbone y afférents, AMEA TOGO SOLAR et ELEMENTSIX souhaiteraient organiser une consultation publique avec toutes les parties prenantes de la communauté locale de Bliha afin d'échanger sur un certain nombre de sujets relatifs aux impacts du changement climatique.

En effet, le projet contribuera à l'amélioration de la résilience de la région Bliha afin de réduire les coûts et les perturbations liés aux impacts du changement climatique. La résilience nécessite de prendre en compte les menaces, les dangers, les besoins et les capacités de l'environnement, d'où la nécessité de recueillir vos témoignages et vos expériences vis-à-vis de la centrale solaire. Au cours de la session, vous serez invités à partager vos témoignages, questions et opinions sur les impacts sociaux et environnementaux du projet.

Les avantages de votre participation à la réunion seront les suivants :

- Une meilleure compréhension du projet et de ses impacts environnementaux et sociaux.
- Des explications sur les causes et les conséquences du changement climatique.
- La nécessité de mettre en œuvre des activités d'atténuation et la manière dont elles peuvent donner lieu à des crédits carbone.
- L'explication du fonctionnement de VCS/VERRA et de la manière de délivrer des crédits carbone crédibles.
- Une occasion de s'assurer que les plans d'atténuation des projets tiennent compte des vulnérabilités de la région.

Par la présente, nous avons le plaisir de vous inviter, vous et vos administrés (parties prenantes de la communauté locale de Bliha), à participer à la réunion de consultation publique qui se tiendra le 5 juillet 2022 au centre de conférence international de Bliha à partir de 12h00. Cette consultation sera conduite conformément aux exigences de la norme VERRA-VCS.

En vous remerciant vivement de votre assistance, je vous prie d'agréer, Monsieur le **Préfet**, mes salutations les plus distinguées

Joel Musikigala
Directeur Général Adjoint
AMEA Togo Solar SAU

Copie pour information :

Directeur **Adjoint**, Agence togolaise d'électrification rurale et des énergies renouvelables (ATZER)

AMEA Togo Solar SAU

Société Anonyme Unipersonnelle avec Conseil d'Administration
Capital : 10 000 000 de FCFA
Siège social : Lomé, 27, Rue **0000** quartier des Etoiles, 05 BP : 92210
RCRA : TG-COM/2019 B 2119
NIF : 1001820081

The grievance mechanism for the project will comply with the following principles:

- Clarify at the outset what is the purpose of the procedure;
- Assure people that there will be neither cost nor retribution associated with lodging a grievance; and
- The entire process (i.e. how a complaint is received and reviewed, how decisions are made and what possibilities may exist for appeal) will be made as transparent as possible by putting it into written form, publicizing it and explaining it to the stakeholders.

To properly manage the potential impacts identified in the ESIA, a grievance mechanism based on the appointment of a Community Liaison officer in charge of managing community complaints has been designed.

The methods to access the grievance mechanism have been disclosed to the local communities during the ESIA consultations. The grievance mechanism will be publicized at the site gate, at the Local Community Chief's Offices, and at the Project Company's website. All information about grievance procedures and grievance registers kept on site will be in English, but grievance forms can be filled in French.

However, since the project-affected communities have no or minimal access to the internet, but the great majority of them have a mobile phone, it's expected that a number of grievances will be received by phone. For such a reason the grievance mechanisms will provide a phone number that will be made available to the communities. Access to the mechanism will be free of cost. No costs nor retribution will be associated with lodging a grievance.

In addition to the explanation provided verbally, the Community Liaison Officer will provide the communities with printed materials (e.g., posters) to be affixed in prominent areas, providing the key facts about the mechanism and contact information, consisting in a dedicated phone number. Communication and awareness raising activities regarding the Grievance Mechanism will be repeated at regular intervals.

2.3 Environmental Impact

An Environmental and Social Impact Assessment (ESIA) was carried out for the project. The activities during construction and operation of the project have been analyzed to identify correspondent environmental impacts and assess their significance. The summary of the findings of the report was as follows:

Impact Assessment methodology

The following assessment of impacts distinguishes between the construction, operation, and decommissioning phases.

- Land use
- Biodiversity
- Air quality and noise levels
- Water resources
- Traffic and Transport
- Existing infrastructure
- Visual impacts
- Socioeconomic impacts

Land, land use, and waste management

As the land consists of trees and agricultural activities, there might be some changes in the soil properties resulting from the soil replacement for the foundations and from construction of the plant intermediate road. Such impacts could only affect the subsequent use of the project site after decommissioning and would be mitigated through the site preparation works. There might also be some impacts on the soil at the project site resulting from accidental leaks of waste or oils used by the construction machinery. However, such spills are expected to be limited through the Spill Prevention Plans and having spill kits available throughout the site and the Hazardous Waste Management plan.

Impacts on land beyond the footprint of the project will also include domestic solid waste in addition to some construction wastes such as plastics, metals, wood, etc. Because the quantity of such waste is expected to be limited, the correspondent impacts on waste disposal sites are minor.

Air quality

The air emissions expected during the construction of the project were the airborne dust resulting from earthworks, in addition to the emissions of different construction machinery such as power generators, excavators, loaders, vehicles, etc. In the absence of sensitive receptors in the area, emissions during construction, such as exhaust gases have low significant impact, and may only temporarily affect users, as the average speed on the road is quite high, during passing the area in addition to slight impacts on the construction workers.

Mitigation measures such as spraying loose soil during excavation and filling operation can be adopted to minimize airborne dust.

Noise levels

Normally construction works include noisy activities related to the operation of construction machines, possible hammering, drilling works, soil compaction in addition to the noise generated from construction related trucks. The impacts of construction noise are likely to cause some considerable effects on neighbors as the neighboring areas are closely populated.

The contractor was encouraged to minimize noise emissions, such as turning off engines of unused equipment, use lower noise emission equipment to the extent possible and use silencer for noisy equipment (such as generators) to comply with the local noise regulations.

Water resources

During construction, temporary movable construction yard (for storage of materials and servicing of machinery) and temporary offices, for construction staff, were erected at a place. The area included simple sanitary facilities. Water supply was via tankers from the central pipeline, which will normally cover domestic uses, uses for soil spraying and uses for mixing concrete. The amount of water consumed in construction sites is expected to be relatively little amounts, in

which the quantitative impacts on water resources are negligible. The mitigation measures include collecting wastewater in waterproof collection tanks and evacuating it frequently to the nearest wastewater treatment plant or sewerage system.

Traffic and Transport

During the construction, heavy transport was used to transfer PV modules and other electric equipment to the construction sites. Since the road networks have low levels of service, truck convoys carrying equipment to the site may cause bottlenecks and impact the current traffic flow.

A plan to develop traffic management was introduced to ensure equipment is delivered in a timely fashion and in good condition, whilst complying with the local legislation.

2.4 Public Comments

No comments received during the public comment period.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The methodology applied is:

- ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0

The following tools referred by the methodology have been used and their applicability is discussed in section 3.2.

- TOOL01: Tool for the demonstration and assessment of additionality --- Version 7.0.0

TOOL07: Tool to calculate the emission factor for an electricity system --- Version 7.0The following tools have also been applied to show additionality:

- TOOL24: Common practice --- Version 3.1
- TOOL27: Investment analysis --- Version 9.0

3.2 Applicability of Methodology

The project activity is a solar project with a total installed capacity of 70 MW. As such, the ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0 is

applicable for this project. The project activity involves the installation of a new power plant at a site where no renewable power plant previously existed and as such, is a Greenfield Power Plant.

The methodology is applicable under the following conditions:

Methodological Applicability	Project Justification
This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units; (d) Involve a rehabilitation of (an) existing plant(s)/unit(s);or (e) Involve a replacement of (an) existing plant(s)/unit(s).	Option (a) - The project is a grid-connected solar energy power generation project that is a greenfield installation.
The methodology is applicable under the following conditions: a) The project activity may include renewable energy power plant/unit of one of the following types: hydropower plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and	Option (a) – The project activity is a renewable energy (solar power) power plant.

defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	
In case of hydro power plants, one of the following conditions shall apply: a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4W/m²; or d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4W/m², all of the following conditions shall apply: i. The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; ii. Water flow between reservoirs is not used by any other	Not applicable. This project is not a hydro power plant.

hydropower unit which is not a part of the project activity; iii. Installed capacity of the power plant(s) with power density lower than or equal to 4W/m² shall be: a) Lower than or equal to 15MW; and b) Less than 10 per cent of the total installed capacity of integrated hydro power project.	
In the case of integrated hydro power projects, project proponent shall: a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if	Not applicable. This project is not an integrated hydro power plant.

any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
The methodology is not applicable to: a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; b) Biomass fired power plants/units.	Not applicable. This project activity does not involve switching of fossil fuels to renewable energy or biomass fired power plants.
In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	Not applicable. This project activity is not a retrofit, rehabilitation, replacement, or capacity addition.
In addition, the applicability conditions included in the tools referred to below apply.	See below tables.

TOOL01: Tool for the demonstration and assessment of additionality --- Version 7.0.0 is applicable under the following conditions:

Tool Applicability	Project Justification
The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They	Not applicable. No new methodology is proposed.

may also submit revisions to approved methodologies using the additionality tool.	
Once the additionality tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	The additionality tool is applied, hence applicable.

TOOL07 “Tool to calculate the emission factor for an electricity system” version 7.0 is applicable under the following conditions:

Tool Applicability	Project Justification
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	The project is a grid-connected solar energy power generation project and thus substitutes grid electricity.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e., option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in	Steps involved in calculation of Emission Factor is included in section 3.1 of the VCS PD as per the required tool.

generation and not to other aspects such as transmission capacity.	
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The Project is located in non-Annex I country and hence the tool is applicable.
Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	Not applicable. No biofuel is involved in this project activity.

TOOL24 “Common practice” Version 3.1 is applicable under the following conditions:

Tool applicability	Project Justification
This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	“Tool for the demonstration and assessment of additionality” has been applied. Hence, the tool is applicable.
In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Not applicable. No approach that are different from those described in the tool has been used.

TOOL27 “Investment analysis” Version 9.0 is applicable under the following conditions:

Tool applicability	Project Justification
This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the	“Tool for the demonstration and assessment of additionality” has been applied. Hence, the tool is applicable.

guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	
In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Not applicable. No approach that are different from those described in the tool has been used.

3.3 Project Boundary

The spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the solar power plant is connected to.

The electricity system is defined by the spatial extent of the power plants that are physically connected through transmission and distribution lines to the solar power plant and that are covered by either single or layered dispatch area.

Thus, the project boundary for this activity is the national/sub-national grid in Togo.

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Included	Main emission source
	CH ₄	Excluded	Minor emission source
	N ₂ O	Excluded	Minor emission source
Project	CO ₂	Excluded	Minor emission source

Source	Gas	Included?	Justification/Explanation
CO ₂ emissions from combustion of fossil fuels for electricity generation	CH ₄	Excluded	Minor emission source
	N ₂ O	Excluded	Minor emission source

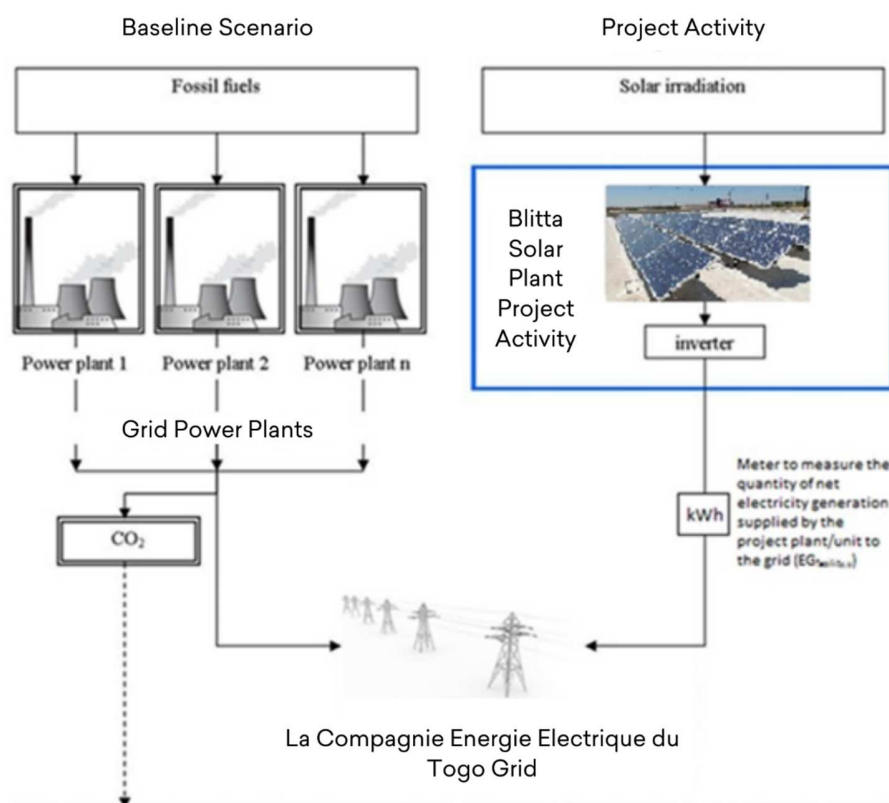


Figure 2 - Diagram of project boundary

3.4 Baseline Scenario

According to the ACM0002: Grid-connected electricity generation from renewable sources — Version 20.0 methodology. “If the project activity is the installation of a Greenfield power plant, the baseline scenario is the electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in “TOOL07: Tool to calculate the emission factor for an electricity system.”

The project activity is the installation of a new power plant at the site and is not a capacity addition or retrofit or replacement of any other existing plant. Any electricity generated and exported from the project activity would in its absence have been generated in the grid dominated by fossil fueled power plants, which represents the baseline scenario. The project scenario is a Solar photovoltaic power plant that would generate energy and supply it to the national grid through La Compagnie Energie Electrique du Togo⁷.

The baseline, the scenario existing prior to the implementation of the project activity, is the continuation of the current practice. The electricity generated by the project activity would have been generated by the addition of new fossil fuel-based generation sources to the grid to supply the additional electricity capacity required.

3.5 Additionality

The project activity generates renewable electricity by means of solar PV technology, which is a renewable, zero emission source of energy. Furthermore, it is important to note that the project activity is not mandated by any laws or statutes in Togo. Additionality of the project activity is demonstrated through a project specific additionality test.

Baseline considerations for the project are based on approved consolidated baseline methodology ACM0002 (Version 20.0). The methodology requires the project proponent to determine the additionality based on the “CDM - Methodological Tool “Tool for the demonstration and assessment of additionality”, Version 7.0.0”. The tool provides for a stepwise approach to demonstrate and assess the additionality of a project. These steps are: (a) Step 0 Demonstration whether the proposed project activity is the first-of-its-kind; (b) Step 1 Identification of alternatives to the project activity; (c) Step 2 Investment analysis; and (e) Step 4 Common practice analysis.

The stepwise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:

Step 0: Demonstration whether the project activity is the first-of-its-kind

According to the CDM – Methodological Tool “Tool for the demonstration and assessment of additionality”, Version 7.0.0”, this step is optional. Therefore, no analysis or evidence will be provided under this step. The project proponent will prove the additionality starting from Step 1 below.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity- Identify realistic and credible alternative(s) available to the project participants or similar project developers that provide outputs or services comparable with the proposed CDM project activity.

⁷<https://energie.gouv.tg/projet/construction-dune-centrale-solaire-de-50-mwc-a-blitta/>

The project activity is generating electrical energy using renewable energy resources that are fed into the national electricity grid. Hence, the following alternatives are considered:

Alternative 1: The project activity undertaken without being registered under VERRA.

The option of project activity being undertaken without being registered under VERRA is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, this is not a plausible option as deployment of the project activity faces significant financial barriers established in the subsequent section. The electricity generated from the project activity would have been fed to the Togo national grid, which has existing and new fossil fuel-based power plants that would have been used in the absence of the project activity.

Alternative 2: Continuation of the current situation (No project activity and the equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid)

In such case the project proponent would have withdrawn from an investment in the project activity and continued with its usual business activities. In such case the grid would have continued with the fossil fuel-based power plants resulting in GHG emissions. Hence, existing power plants and new capacity add-ons from fossil fuel-based power plants can be considered as an appropriate, realistic & credible baseline alternative for the project activity.

Outcome of Sub-step 1.a: Identified realistic and credible alternative scenario(s) to the project activity.

Of the two alternatives outlined above, the first alternative is not realistic as project activity as it is not viable without the benefits of carbon credits. The second alternative on the other hand is the baseline scenario for the project activity as per methodology.

Therefore, the continuation of the current situation is the most plausible alternative to the project activity. The project being a green field project activity and the baseline scenario, in line with the methodology, is “the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

Sub-step 1b: The alternative(s) shall be in compliance with all mandatory applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g., to mitigate local air pollution.

The implementation of the project activity is consistent with mandatory laws and regulations in Togo and relevant clearances and approvals have been obtained.

Outcome of Sub-step 1.b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account

the enforcement in the region or country decisions on national and/or sectoral policies and regulations.

Both the alternatives identified above are found to be in compliance with the mandatory laws and regulations taking into account the enforcement of the legislations in the region/country and relevant decisions on national and/or sectoral policies and regulations.

However, Alternative 2 “Continuation of the current situation” has been selected as the appropriate baseline alternative for this project activity in line with the methodology.

Step 2: Investment analysis

An investment analysis is carried out to determine whether the project activity is economically or financially less attractive than at least one other alternative, identified in step 1, without the revenue from the sale of carbon credits. This is demonstrated in line with the tools for the relevant sections as per “Investment Analysis” (Ver 9.0).

Sub-step 2a: Determine appropriate analysis method

The project activity envisages the export of renewable energy to the Togo electricity grid and the revenues from the sale of electricity are based on the terms and conditions within the Power Purchase Agreement (PPA).

A simple cost analysis (Option I) cannot be used as the analysis method as the sale of electricity will result in a revenue stream during the project operation. In the absence of the project activity, the continued usage of the grid electricity would have been the most plausible option, which does not require an investment.

Hence an investment comparison analysis (Option II) is also not appropriate for the project activity.

After eliminating Option, I and Option II, the use of a Benchmark analysis (Option III) is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project activity for the investors, as well as provides a measure of the viability of the investment to generate revenues during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be applied in context of the project activity.

Sub-step 2b (Option III): Apply benchmark analysis

The investment analysis using the Benchmark analysis approach (Option III) has been chosen.

Further, this method illustrates that the assessment of the project by the project proponent was carried out prior to the decision to undertake the project was taken and management approval had been granted.

Choice of Financial Indicator:

According to the “Tool for demonstration and assessment of Additionality” Version 07.0.0, the financial indicator can be based either on (1) project IRR or (2) equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for the analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for the project activity has been chosen as post tax equity IRR.

According to Methodological Tool27: Investment Analysis Version 9.0, the Default Value Benchmark of Equity IRR in Togo under Group 1, Page 15 (Energy Industries, Energy Distribution, Energy Demand, Waste Handling and Disposal) is 16.25%⁸. Version 9.0 of Tool27 is used as it was valid when the investment decision was taken, which was in March 2019 according to the Memorandum of Understanding.

As the investment analysis is carried out in nominal terms, the benchmark value must also be taken in nominal terms. The nominal benchmark can be calculated using the below formula:

$$\text{Nominal benchmark} = \{(1 + \text{Real benchmark}) \times (1 + \text{Inflation rate})\} - 1$$

Where:

$$\text{Real benchmark} = 16.25\%$$

Inflation rate = 3.2% - As there was no forecasted inflation data available from the Central Bank of West African States (BCEAO) of which Togo is a member, the International Monetary Fund was the reference used. The average forecasted inflation rate for Togo for the first 5 years of the crediting period is 3.2%⁹. Hence, the nominal benchmark value is calculated to be 19.97%

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III):

The post-tax Equity IRR is evaluated for the entire lifetime of the project activity, i.e., 25 years. It is calculated based on the cash outflows from and cash inflows into the project activity as shown in the provided spreadsheets.

The IRR and Benchmark analysis shows that the project investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark). Thus, it can be concluded that the project activity is not the business-as-usual scenario and thus is additional.

Post-Tax Equity IRR	Nominal Benchmark IRR
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<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v9.0.pdf>

⁹ <https://www.imf.org/en/Countries/TGO#whatsnew>

7.98%	19.97%
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Sub-step 2d: Sensitivity Analysis

The objective of the sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the scenario presented, in order to provide a cross-check on the robustness of the assumptions used in the development of the investment analysis

The following factor has been subjected to the sensitivity analysis:

1. O&M cost
2. PLF (Plant Load Factor)
3. Tariff Rate

The results of the sensitivity analysis show that even with a variation of +10%/-10% in PLF, O&M costs and Tariff Rate, the Equity IRR is significantly lower than the benchmark IRR. And it is evident from the results given above that the project activity remains additional even under the most favourable conditions.

Sensitivity analysis	Post-Tax Equity IRR			
Variation %	-10%	Normal	+10%	Nominal Benchmark
O&M costs	8.59%	7.98%	7.36%	19.97%
PLF	4.69%	7.98%	11.40%	19.97%
Tariff Rate	4.69%	7.98%	11.40%	19.97%

The table below shows the values that breach the benchmark for each of the parameter identified:

Parameter	Original Value	Parameter value that breaches the benchmark	% Difference
O&M (USD Mn.)	1.40	0.00*	-100.00%
PLF @ P50 (%)	20.46%	27.34%	33.59%
Tariff Rate (USD/MWh)	69.49	92.83	33.59%

*Even if the O&M Costs drop by 100%, the equity IRR becomes 14.07% which is still below the benchmark value.

Step 4: Common practice analysis

As per the "Tool for demonstration and assessment of additionality" (Version 07.0.0), Step 2 Investment Analysis shall be complemented with an analysis of the extent to which the project

type (e.g., technology or practice) has already diffused in the relevant sector and region within the country. This test is a credibility check to complement the investment analysis. The Methodological Tool followed in this Analysis is Methodological Tool24: Common Practice, Version 3.1.

Common Practice Analysis:

Step (1): Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

Range	Capacity (incl. phase 2)	Unit
+50%	105	MW
Project activity capacity	70	MW
-50%	35	MW

Step (2): Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

- the projects are located in the applicable geographical area
- the projects apply the same measure as the proposed project activity
- the projects use the same energy source/fuel and feedstock as the project activity, if a technology switch measure is implemented by the project activity
- the projects implemented produce goods or services with comparable quality, properties and applications areas (e.g., clinker) as the project plant
- the capacity or output of the projects is within the applicable capacity or output range calculated in
- the projects started commercial operation before the project documents have been published for global stakeholder consultation or before the start date of the project activity, whichever is earlier for the proposed project activity.

Analysis of Step 2

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:

- as the project is located in Togo, projects in the geographical area of Togo have been chosen for the analysis
- the project activity is a green-field solar power project and uses measure (b) “Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies”. Therefore, projects applying the same measure (b) are candidates for similar projects

- the energy source used by the project activity is solar. Hence, only solar energy projects have been considered for the analysis
- the project activity produces electricity; therefore, all power plants that produce electricity are candidates for similar projects
- the capacity range of the projects is within the applicable capacity range from 35 MW to 105 MW
- as per the methodological tool the projects that have started commercial operation before the project documents have been published for global stakeholder consultation or before the start date of the project activity, are to be selected for consideration

The findings of the analysis of Step 2

There are no similar projects in Togo that fulfil above-mentioned conditions. The Blitta-Togo powerplant is considered as the first renewable energy facility in the country by International Renewable Energy Agency¹⁰.

Step (3): Within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing project verification.

$N_{all} = 0$

Step (4): Within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity.

$N_{all} = 0$

Step (5): Calculate factor $F = 1 - N_{diff} / N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the project activity that deliver the same output or capacity as the project activity.

$N_{all} = 0$

Conclusion

¹⁰ <https://www.irena.org/newsroom/pressreleases/2021/Jun/Togo-Inaugurates-50MW-Solar-Plant-Financed-Under-IRENA-ADFD-Facility>

Based on the above analysis it can be concluded that the project activity is not a common practice and thus is considered additional.

3.6 Methodology Deviations

Not applicable – there are no deviations.

4 QUANTIFICATION OF EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

According to the latest version of the methodology, ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0, Section 5.5, Paragraph 47, since the project activity is the installation of Greenfield Renewable Energy Power Plant, the baseline scenario is the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in “TOOL07: Tool to calculate the emission factor for an electricity system”.

The scope and applicability of Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool, covers the grid emission factors for the electricity system of the WAPP (West African Power Pool) Version 01.0.

This Standardized Baseline can be used under the following conditions:

Baseline Procedure	Justification
The project activity is implemented in any one of the WAPP member countries i.e., Benin, Burkina Faso, Côte d'Ivoire, Ghana, Mali, Niger, Nigeria, Senegal and <u>Togo</u> and is connected to the WAPP	The project activity is in <u>Togo</u> and is connected to the WAPP.
The CDM approved methodology that is applied to the project activity requires the determination of CO ₂ emission factor(s) through the application of the grid tool	ACM0002 requires the determination of CO ₂ emission factor(s) through the application of the grid tool “TOOL07: Tool to calculate the emission factor for an electricity system

The project activity uses the ex-ante options for both the operating margin and build margin grid emissions factors, as described in the grid tool, and therefore no monitoring or recalculation of the emission factor during the crediting period is required	The project activity uses the ex-ante options for both the operating margin and build margin grid emission factors, as described in the grid tool. Therefore, no monitoring or recalculation of the emission factor during the crediting period is required.
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According to the ACM0002, Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel-fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above the baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants.

The baseline emissions are to be calculated as follows:

$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$ Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of "TOOL07: Tool to calculate the emission factor for an electricity system" (tCO₂/MWh) referring to the Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool, Version 01.0.

Calculation of $EG_{PJ,y}$

In case of Greenfield power plant development, the calculation of the electricity system's combined margin (CM) shall be based on the following equation, according to the methodology:

$EG_{PJ,y} = EG_{facility,y}$ Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Calculation of $EF_{grid,CM,y}$

As stated before, the “TOOL07: Tool to calculate the emission factor for an electricity system” allows referring to the Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool, Version 01.0.

The ASB0034-2021 Standardized Baseline states that for wind and solar power generation project activities, the $EF_{grid,CM,y}$ to be used is 0.573 tCO₂/MWh.

4.2 Project Emissions

The project activity involves renewable power generation. Therefore, $PE_y = 0$

4.3 Leakage

No leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g., extraction, processing, transport etc.) are not calculated.

4.4 Net GHG Emission Reductions and Removals

Baseline Emissions:

Baseline Emissions are calculated as follows:

$$BE_y = EG_{pj,y} * EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂ /year)

$EG_{pj,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y. The value is assumed based on the Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh).

The total estimated quantity of net electricity supplied to the grid:

$$EG_{pj,y} = 107,027 \text{ MWh/yr}$$

The Grid Emission Factor is:

$$EF_{grid,CM,y} = 0.573 \text{ tCO}_2/\text{MWh}$$

The calculation of the baseline emissions of the project activity is:

$$BE_y = 107,027 \text{ MWh/yr} * 0.573 \text{ tCO}_2/\text{MWh} = 61,326 \text{ tCO}_2/\text{yr}.$$

Leakage:

Leakage emissions are not applicable.

Project Emissions:

Project Emissions are as shown below:

$$PE_y = 0 \text{ tCO}_2\text{e/yr}$$

Emission Reductions:

$$ER_y = BE_y - PE_y$$

$$ER_y = 61,326 \text{ tCO}_2/\text{yr} - 0 \text{ tCO}_2\text{e/yr} = 61,326 \text{ tCO}_2\text{e/yr}$$

A deration factor of 0.50% has been applied for every year of the project lifetime while calculating the net annual generation.

Year	Estimated baseline emissions or removals (tCO ₂)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
Year 2021	39,985	0	0	39,985
Year 2022	70,822	0	0	70,822
Year 2023	70,468	0	0	70,468
Year 2024	70,308	0	0	70,308
Year 2025	69,765	0	0	69,765
Year 2026	69,416	0	0	69,416
Year 2027	69,069	0	0	69,069
Year 2028	30,774	0	0	30,774
Total	490,607	0	0	490,607

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid, CM, y}
Data unit	tCO ₂ /MWh
Description	Combined Margin CO ₂ grid emission factor in the year y
Source of data	Standardized Baseline ASB0034-2021
Value applied	0.573 tCO ₂ /MWh
Justification of choice of data or description of measurement methods and procedures applied	The availability of a Standardized Baseline for the Grid Emission Factor for West African Power Pool Version 01.0
Purpose of Data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period

5.2 Data and Parameters Monitored

Data / Parameter	EG PJ,y				
Data unit	MWh/yr				
Description	Quantity of net electricity generation supplied to the grid in year y by the solar plant				
Source of data	Meter readings				
Description of measurement methods and procedures to be applied	Net electricity generation will be electronically metered and stored under the Plant Manager’s authority				
Frequency of monitoring/recording	Continuous measurement and monthly recordings				
Value applied	107,027				
Monitoring equipment					
		Meter Description	Serial Number	Accuracy Class	Make
	1	Power Trafo.1 Main	56198213	0.2S	Landis+Gyr

	2	Power Trafo.1 Check	56198214	0.2S	Landis+Gyr
	3	Power Trafo.2 Main	56198215	0.2S	Landis+Gyr
	4	Power Trafo.2 Check	56198216	0.2S	Landis+Gyr
	5	Blitta feeder Main	56198203	0.2S	Landis+Gyr
	6	Blitta feeder Check	56198217	0.2S	Landis+Gyr
	7	Auxiliary Trafo. Main	56198221	0.2S	Landis+Gyr
	8	Auxiliary Trafo. Check	56198222	0.2S	Landis+Gyr
	QA/QC procedures to be applied Calibration of the meters will be carried out as per manufacturer specifications. Inspection and testing of accuracy will be carried out on annual basis				
Purpose of data		Calculation of baseline emissions			
Calculation method		<i>Not applicable</i>			
Comments		Data will be archived in paper & electronically for a period of 2 years beyond the end of crediting period or of the last issuance of credits for this project activity, whichever occurs later.			

5.3 Monitoring Plan

The proposed Monitoring Plan describes the Management System and Procedures to be developed and implemented by the project owner for this project in order to ensure consistency with the methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0 which is applied in the context of the project activity.

The developed monitoring plan contains the following sections:

1. Monitoring parameters
2. Monitoring Points

3. Monitoring equipment
4. Management structure
5. Monitoring guidelines, procedures and documentation
6. Regular staff training

All data required for the calculation of emission reductions associated with the project activity will be monitored in accordance with the approved methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0 and will be archived electronically and on paper throughout the crediting period and up to 2 years thereafter.

A Monitoring Management and Quality Assurance System (MMQAS) will be developed and implemented to ensure that the results from the project activity are reliable and transparent.

The complete MMQAS, including data and documents from project operation, will be open for review to the VVBs contracted for verification.

Monitoring Parameters

The tables under section 5.1 provide information on the required monitoring parameters as per approved methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0 and referred tools, required for the ex-post calculation of project emissions associated with the project activity.

Monitoring Points

Single line diagram

Monitoring equipment

The physical monitoring system consists of the following meter equipment:

Electricity meter: AMEA SOLAR TOGO SAU to monitor the quantity of net electricity generation supplied to the grid in year y by the solar plant (EG PJ,y)

As shown in the calculations of section 4.4 and according to the methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 20., baseline emissions will be calculated by multiplying the quantity of net electricity generation supplied to the grid in year y by the solar plant (EG PJ,y with) with the grid emission factor in the year y (EF grid, CM, y)

Management structure

A VCS team will be designated to be responsible for carrying out the required monitoring tasks for the project activity. Both management people and technical workers will be present in the VCS team. All of the team members must have passed internal monitoring training.

The management structure consists of 3 levels to guarantee a step-by-step management of the project activity, which will cast concrete responsibilities on the appointed person who will be actually in charge of a specific task.

The 3 levels refer to:

1. Solar Plant Manager
2. VCS project manager
3. VCS staff

Before the start of the crediting period, clear roles and responsibilities will be assigned to all staff involved in the project activity. A VCS Project Manager will be appointed who will be responsible for monitoring activities and data management associated with the project activity. All operators involved in the collection of data and records will be coordinated by the VCS Manager.

Monitoring guidelines, procedures and documentation

The introduced Monitoring Management and Quality Assurance System is building on written guidelines and procedures provided to the operational VCS team staff and to the local VCS management.

The procedures will be developed by the external consultant according to the approved methodology ACM0002: Grid-connected electricity generation from renewable sources -- Version 20.0. All tasks under the monitoring plan will be carried out strictly according to the provided guidelines and procedures.

The Monitoring Management and Quality Assurance System documentation is designed to enable VCS staff members to perform the required tasks following the given guidelines and procedures. Providing documentation designed specifically for the required tasks will strongly benefit availability and transparency of data streams and processes which are relevant for the VCS project.

The VCS Project Manager will carry out frequent performance and data cross-checking, to ensure that staff are following the Monitoring Management and Quality Assurance System.

All documents filled out and data acquired during the project will be archived separately for backup.

Regular Staff Training

The Monitoring Management and Quality Assurance System is implemented and will be maintained by an external consultant. In order to qualify local operational staff to carry out required regular tasks external consultant will provide professional training, both in class and practical application, to the designated VCS team staff before the start of the crediting period and ongoing regular training at least once a year for the duration of the project.

Only trained staff will be members of the VCS team and only trained VCS staff members will be allowed to perform any tasks related to monitoring activities.

The VCS Manager is permitted to train new staff. By this it will be ensured that operational VCS staff are qualified to carry out the relevant tasks under the Monitoring Management and Quality Assurance System.

Based on the annual monitoring results VCS staff will prepare the annual Monitoring Report. An overall AQ/QC on the Monitoring Spreadsheet and Monitoring Report will be carried out by an external Consultant before the documents are submitted to a verifying VVB.

APPENDIX

APPENDIX A: LOCAL STAKEHOLDER CONSULTATION REPORT

Background & Objective

Stakeholders' Consultation

Togo Blitta 50 MW Solar Power Plant Project

Venue: CICB (Centre International de Conférence de Blitta)

Date: Tuesday, 05-July-2022

Time: 08:00 AM GMT

A Local Stakeholder Consultation process has been conducted in accordance with the VERRA- Verified Carbon Standard (VCS) requirements, towards the successful registration of emission reduction projects/programs aiming to generate carbon credits for the project activity.

The government of Togo has inaugurated the largest solar PV plant in West Africa and the first renewable energy facility in the country. This 70 MW solar PV plant facility, located in Blitta, will supply reliable, clean electricity to hundreds of thousands of homes and businesses in the country and save more than one million tonnes of CO₂ emissions throughout its lifetime.

The project activity, which has a lifetime of 25 years, is expected to produce an average of 107,000MWh of electricity annually once the plant is fully operational. The total estimated emission reductions of the project activity are 490,607 tCO₂e during the first crediting period. The ex-ante emission reduction calculation shows that the project will avoid 61,326tCO₂e on average, annually.

The objective of the Stakeholders' Consultation event is to take due diligence of all comments, questions, and opinions in an open and transparent manner towards the projects' economic, social, and environmental impacts, and sustainability. It is also to raise the stakeholders' awareness level about the procedure to register such projects under the VCS.

This report is devised to provide the outcomes of the stakeholder consultation, and to ensure the transparency of the process through documentation.

1. Stakeholders & Mode of Invitations

In this manner, stakeholders mean the public, including individuals, groups or communities relevant to the Project Activity, end-users, and national authorities. Invitations were sent by AMEA SOLAR TOGO SAU to the Prefect (Village Chief), who then distributed them to the following entities (Refer to Appendix A for Invitation & Agenda):

- AMEA SOLAR TOGO SAU
- JAKSON
- Action Sociale
- Togo Police
- Radio Etincelle
- Chefferie
- IEPP Blitta
- NSCT Blitta
- Treasurer of Blitta
- ANADEB Blitta
- Responsable des Blitta – Village
- Chef de Yaloumbe Canton
- Gendarmerie
- Irto Blitta
- Chef Village

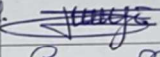
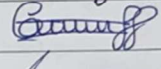
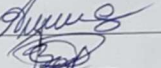
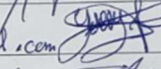
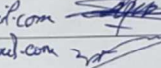
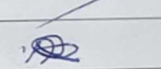
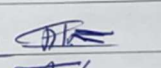
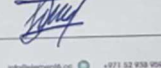
2. Attendees



Stakeholder Consultation Registration

Project: Togo Blitta 50 MW Solar Power Plant Project

Date: July 5th 2022

No.	Name	Organisation	Designation	Email / Phone #	Signature
1	JAO ELOM	JAKSON	HSE & social specialist	elomya202@gmail.com	
2	TCHIRITEMA siliti	Préfecture Trésorerie de Blitta	Trésorier de Blitta	siliti79@gmail.com	
3	ELHYAOUNOU AKOUA	Chef Village	Blitta village	91800544	
4	BILAKE Ekpouwa	IEPP Blitta	Inspection Edu.	90170692	
5	DAZIMWAI Yao B.	Maire Blitta 1	Commune Blitta 1	90872040	
6	YAKE Tommotopi	Radio Etincelle	Radio Blitta	90293830 radioetincelle@gmail.com	
7	SAMBAWE Jeanpapaie	Irto ATOP	Blitta	90141769 pidossambanir@gmail.com	
8	BADABAKE Bakpakou	NSCT Blitta	chef d'usine	pierebadabake@gmail.com 90183142	
9	ASSIATKO Dora	Responsable des Blitta-Village	Blitta-Village	91285969	
10	ASSAKPO Komlan	Président des jeunes	Blitta-Village	91863042 92350399	
11	BOUTOULI T. Solimdon	ANADEB	AAOB/Blitta	90734210@gmail.com	

info@elementsix.co +971 52 938 9590

CTEC, TechnoHub Building 1, DGC Dubai, UAE www.elementsix.co



element six

Stakeholder Consultation Registration

Project: Togo Blitta 50 MW Solar Power Plant Project

Date: July 5th 2022

No.	Name	Organisation	Designation	Email	Phone #	Signature
12	Hervé RIoux	AMEA POWER	Site Representative	herve.rioux@amea-power.com	93343763	
13	Raouf DAMTARE	AMEA POWER	QHSE Manager	raouf.damtare@ameatog solar.com	90021910	
14	KPANAIKE D. Donkor	Actim Sociale	Directeur Préfectoral	flavien.kpanaike@gmail.com	90387637	
15	KAYI Modjo	TIE	Chef d'Agence	raouf.damtare@gmail.com	90017802	
16	GNOAME Ben	Gendarmérie	Représentant du CB-Brigade	gnognoame@gmail.com	90304020	
17	KONDO Fabrice	Radio Etincelle	Journaliste	kondofab@gmail.com	90849830	
18	Lieutenant BÉZRA Kossi	POLICE	Commandant CRI Blitta	kossibezra@gmail.com	91090405	
19	ALOUKEURO Komlan	chef de Canton	chef de Canton		92275527	
20	AKATI LESS OROKINA	chef de Blitta	chef de Village		90750841	
21	NABSONA Mamoudou	Jakson - Togo	Technicien	mamoudou.nabsona@gmail.com	91935890	
22	KONDO Biredey Abdel-Adl	JAKSON-TOGO	Site Engineer	biredey.kondoa@gmail.com	90853610	
23	DIANEYE-FARE E. Nanané	JAKSON-TOGO	Opérateur	dianeeyefare@gmail.com	91912281	
24	AFFO -NGO Elyseu	OPRF	Environnementaliste Responsable	elyseu.affo@gmail.com	91965217	

N°	Name	organisation	Désignation	Email	Signature
25	Alou Atcha Balababadi	chefferie	chef canton	92683016	
26	TCHALIM Ekpouwa	chefferie	chef canton	90969462	
27	ALIOU I. Moumoutou	COTPOLICE	BLITTA	alioumoumoutou@yahoo.com	
28	ALOUONOU YAOVI	chefferie	chef canton	90966831	
29	DOÏSE Medjina	SG. Préfecture	secrétaire	medjina.doise@gmail.com	
30	LATTAH Kossu W.	Assistant DPS	Blitta DPS	91543808	

3. Speakers

Introduction & Overview of the Stakeholder Consultation Process, GHG Emission reduction projects, Verra-VCS, and Question and Answer session have been conducted by Element 6-FZCO.

4. Summary







The project activity was explained in detail including the baseline scenario, methodology, and the contribution of the project activity to Sustainable Development Goals (SDGs). The carbon markets and their trends were also discussed during the session.

(Refer to Appendix C for full presentation)

5. Stakeholder Discussion

Q. No.	Question	Response
1	How is the project activity beneficial in terms of revenue?	The revenue generated for the project activity is divided between AMEA SOLAR TOGO SAU and the Togolese government.
2	What is the goal of consulting stakeholders for such a project?	It is a requirement to consider and address the feedback, positive or negative, of the stakeholders. A report on the local stakeholder consultation report must be prepared and the details of the meetings

		must be mentioned in the project description document. The document will then be validated by a 3 rd party validator. In addition, it is crucial to be transparent and involve the local community in the project.
3	Are you calculating the emission factor?	Based on the annual electricity generated, Grid Emission Factor (GEF) is considered for the calculations. The GEF used for the project activity is the standardized GEF for the West African countries including Togo. The value of the GEF used is 0.573 tCO ₂ /MWh.
4	What is the revenue generated for a tonne of decarbonized carbon?	The cost is decided based on an agreement between the ClimatePartner and AMEA SOLAR TOGO SAU. The cost depends on the location of the project, technology used, demand and supply, and the market value, among many other factors. The location of the project activity plays an important role in determining the credit value. The value of such a credit would be higher in a developing country compared to a developed country due to the added advantage of social impacts.
5	How many credits can be expected if someone wants to invest in an AFOLU project?	The credits that can be achieved from a project activity depend on various factors such as the type of the project, methodology followed, scale etc.
6	How are the safety concerns of people living close to the project activity addressed?	A minimum safety distance must be considered as part of project requirements before it is approved. In addition, no negative impact on human health was observed by the Environmental and Social Impact Assessment.
7	What are the minimum requirements to invest in an AFOLU project?	The potential of emission reduction is assessed based on the scale of the project. A feasibility study of the project activity must be analyzed before further actions are carried out.

Appendix B – Invitation & Agenda

The invitees were sent the invitation below to attend the stakeholders' consultation event. They were also requested to submit their written comments, should they have any, via email to Amer Arafat at info@element6.cc.



Lomé, le 20 juin 2022

Monsieur M. BOUKARI BATOSSA
Préfet de la préfecture de **Blitta**,
BP: 01 **Blitta**, Gare
République du Togo

Référence : BLT_009_06_22/PREFET/AMEA

Concerne : Projet AMEA - Centrale Solaire de 70 MWc à **Blitta**.

Objet : Invitation des parties prenantes locaux à la consultation publique

Monsieur le **Préfet**,

Dans le cadre de la valorisation du projet de centrale solaire de **Blitta** pour l'obtention des crédits carbone y afférents, AMEA TOGO SOLAR et ELEMENTSIX souhaiteraient organiser une consultation publique avec toutes les parties prenantes de la communauté locale de **Blitta** afin d'échanger sur un certain nombre de sujets relatifs aux impacts du changement climatique.

En effet, le projet contribuera à l'amélioration de la résilience de la région **Blitta** afin de réduire les coûts et les perturbations liés aux impacts du changement climatique. La résilience nécessite de prendre en compte les menaces, les dangers, les besoins et les capacités de l'environnement, d'où la nécessité de recueillir vos témoignages et vos expériences vis-à-vis de la centrale solaire. Au cours de la session, vous serez invités à partager vos témoignages, questions et opinions sur les impacts sociaux et environnementaux du projet.

Les avantages de votre participation à la réunion seront les suivants :

- Une meilleure compréhension du projet et de ses impacts environnementaux et sociaux.
- Des explications sur les causes et les conséquences du changement climatique.
- La nécessité de mettre en œuvre des activités d'atténuation et la manière dont elles peuvent donner lieu à des crédits carbone.
- L'explication du fonctionnement de VCS/VERRA et de la manière de délivrer des crédits carbone crédibles.
- Une occasion de s'assurer que les plans d'atténuation des projets tiennent compte des vulnérabilités de la région.

Par la présente, nous avons le plaisir de vous inviter, vous et vos administrés (parties prenantes de la communauté locale de **Blitta**), à participer à la réunion de consultation publique qui se tiendra le 5 juillet 2022 au centre de conférence international de **Blitta** à partir de 12h00. Cette consultation sera conduite conformément aux exigences de la norme VERRA-VCS.

En vous remerciant vivement de votre assistance, je vous prie d'agréer, Monsieur le **Préfet**, mes salutations les plus distinguées

Joel Musikangala
Directeur Général Adjoint
AMEA Togo Solar SAU

Copie pour information :

Directeur **Général**, Agence togolaise d'électrification rurale et des énergies renouvelables (ATZER)

AMEA Togo Solar SAU

Société Anonyme Unipersonnelle avec Conseil d'Administration
Capital : 10 000 000 de FCFA,
Siège social : Lomé, 27, Rue **Wou** quartier des Etalés, 06 BP - 02210
Blitta, TOGO-LOM 2019 B 2119
NIE : 1001620051

The event's agenda is shown below:

The slide has a background image of a forest with sunlight filtering through the trees. In the top left corner, there is a small green circular logo with a stylized 'V' shape. The title 'TODAY'S AGENDA' is prominently displayed in the upper left. The agenda items are listed in two columns: times on the left and activities on the right.

TODAY'S AGENDA	
08:00 – 09:00	Registration
09:00 – 09:30	Climate Change and VERRA VCS
09:30 – 10:00	The Project Activity
10:00 – 10:30	Q&A and Closing
10:45	End

Appendix C – LSC Presentation



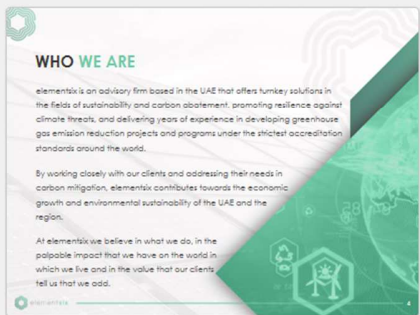
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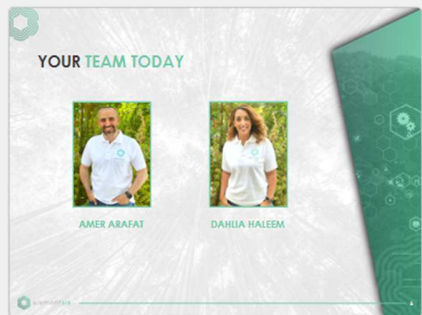
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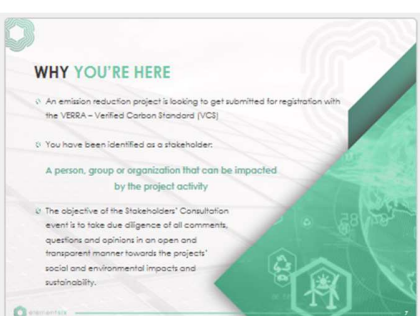
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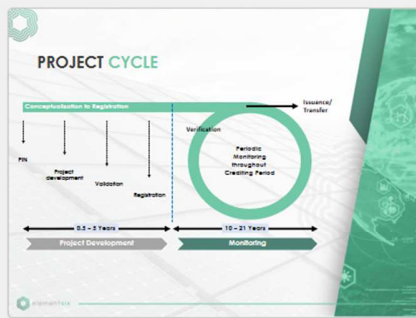
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PROJECT DESCRIPTION

The project consists of the development and implementation of a utility scale solar PV plant to generate and supply renewable electricity to the national grid through La Compagnie Électrique du Togo.

- Capacity: 50-70 MW
- Generation: 88,154 - 123,384 MWh/yr
- Emission reductions: 44,000 tCO₂/yr
- Location: Bitta, Togo
- Commissioning: June 2021
- Uplifts: 23 uplifts
- Owner: AMBA Power LLC
- Other parties: ClimatePartners, elementSix

The project will advance the Togolese national clean energy strategy to increase the share of renewable energy mix to 30% by 2023 and 100% by 2030.

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BASELINE SCENARIO

The baseline scenario is the electricity generated and delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

EMISSION REDUCTIONS

Year	Estimated GHG emission reductions (tCO ₂ e)
2021	23,617
2022	34,496
2023	76,700
2024	76,700
2025	76,700
2026	76,700
2027	76,700
2028	35,827
Total estimated GHG	444,040
Total number of crediting years	7
Average annual GHG	64,231

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METHODOLOGY

The project activity uses approved CDM Methodology:

ACM 0002 - Large-scale Consolidated Methodology, Grid-connected electricity generation from renewable sources (Version 20.0)

ADDITIONALITY

"Additionality testing aims to ensure that only mitigation outcomes that require carbon market support are created under baseline-and-credit schemes, so that carbon finance is not spent on mitigation that would have been achieved anyway."

How to prove additionality:

- The Pro-Share
- Positive Lists of Technologies
- Investment Analysis/Bank Analysis/Common Practice

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SUST. DEVELOPMENT CONTRIBUTION

SDG 3: Good Health and Wellbeing

- Construction of a clinic in Bitta for curative and preventive service
- Distributed essential food and hygiene products to 2,200 families as a response to the lockdown in Togo

SDG 4: Quality Education

- Construction of 3 primary schools (Bitta, Tamegbe and Kossongbe) and renovation of 4 primary schools (Bitta Centre, Comébour, Rinade and Tchongbe)
- Provision of the necessary furniture and educational equipment

SDG 6: Clean Water and Sanitation

- Borehole drilling for water
- Installation of water pump at local primary school

SDG 7: Affordable and Clean Energy

- Installation free electrification of seven primary schools via solar power.

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NO NET HARM

- Monitoring of all activities during construction, operation, (and decommissioning)
- Identification and assessment of environmental and social impacts
- Identification of mitigation measures in Environmental and Social Management Plan (ESMP)
- Socioeconomic benefits outweigh the potential negative environmental and social impacts
- Possible impacts of decommissioning phase have been analysed in the ESIA

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ENVIRONMENTAL IMPACT

- An Environmental and Social Impact Assessment (ESIA) has been carried out for the project
- Activities during construction and operation of the project are and have been designed to identify correspondent environmental impacts and assess their significance
- The following impacts have been assessed:
 - Land use
 - Biodiversity
 - Air quality and noise levels
 - Water resources
 - Traffic and Transport
 - Existing infrastructure
 - Visual impacts
 - Socioeconomic impacts

20

ANY QUESTIONS?

21

THANK YOU

E: info@elementSix.co
W: www.elementSix.co

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Dubai Silicon Oasis, Dubai, UAE

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